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Real-world validation of a multimodal LLM-powered pipeline for high-accuracy clinical trial patient matching

Anatole Callies ¹✉, Quentin Bodinier ¹, Philippe Ravaud ^{1,2,3,4} & Kourosh Davarpanah^{1,4}

Abstract

Background Patient recruitment in clinical trials is hindered by complex eligibility criteria and labor-intensive chart reviews. Prior research using text-only models has struggled to address this problem in a reliable and scalable way due to (1) limited reasoning capabilities, (2) information loss from converting visual records to text, and (3) lack of a generic EHR integration to extract patient data.

Methods We introduce a broadly applicable, integration-free, LLM-powered pipeline that automates patient-trial matching using unprocessed documents extracted from EHRs. Our approach leverages (1) the new reasoning-LLM paradigm, enabling the assessment of even the most complex criteria, (2) the visual capabilities of the latest LLMs to interpret medical records without lossy image-to-text conversions, and (3) multimodal embeddings for efficient medical record search. The pipeline was validated on the n2c2 2018 cohort selection dataset (288 diabetic patients) and a real-world dataset composed of 485 patients from 30 different sites matched against 36 diverse trials.

Results On the n2c2 dataset, our method introduces a new state-of-the-art criterion-level accuracy of 93%. In real-world trials, the pipeline yielded an accuracy of 87%, undermined by the difficulty of replicating human decision-making when medical records lack sufficient information. Nevertheless, users were able to review overall eligibility in under 9 minutes per patient on average, representing an 80% improvement over traditional manual chart reviews.

Conclusions This pipeline demonstrates robust performance in clinical trial patient matching without requiring custom integration with site systems or trial-specific tailoring, thereby enabling scalable deployment across sites seeking to leverage AI for patient matching.

Plain language summary

Recruiting patients for clinical trials is time-consuming and resource-intensive because eligibility rules are complex and medical records are lengthy. We built an artificial intelligence (AI) system that helps match patients to trials by reading both text and images from medical records, including scans, tables, and handwriting. This digital tool finds the most relevant pages, checks each rule step by step, and clearly flags when medical information is missing. We evaluated it on a widely used public dataset and in real clinics across many sites and trials. The system produced reliable, high-quality eligibility assessments, and coordinators were able to review each patient in under nine minutes on average, much faster than manual chart review. Because it works without custom connections to hospital software, it can be deployed broadly to reduce delays and help more patients access studies and new treatments.

Patient recruitment remains a critical bottleneck in clinical development pipelines, with enrollment challenges alone accounting for 30% of phase III trial terminations and substantially prolonging drug development timelines¹. At the core of this problem is the laborious task of identifying suitable patients, a step often complicated by increasingly stringent requirements for trial eligibility. Over the last two decades, the median number of eligibility criteria has surged by 58%, rising from 31 in 2001–2005 to 49 in 2016–2020¹. This added complexity is particularly prevalent in fields such as oncology, neurology, and infectious diseases, where the criteria

counts can exceed 60¹, placing even greater demands on Clinical Research Coordinators (abbreviated as CRCs in the rest of the document). As a result, the pre-screening phase alone can consume up to 60% of total patient recruitment effort². When done manually, each patient pre-screening takes on average 50 minutes in phase III trials. Because as many as 88% of them ultimately fail to meet the criteria, it takes on average more than seven hours to find one eligible patient for Phase III trials².

Prior research has tackled this challenge through two main approaches. First, a “patient-centric” approach to find the relevant trials for a given

¹Inato, Paris, France. ²Center for Research in Epidemiology and Statistics (CRESS), Université Paris Cité and Université Sorbonne Paris Nord, INSERM, INRAE, F-75004 Paris, France. ³Centre d'épidémiologie clinique, AP-HP, Hôpital Hôtel Dieu, F-75004 Paris, France. ⁴These authors jointly supervised this work: Philippe Ravaud, Kourosh Davarpanah. ✉e-mail: anatole.callies@inato.com

patient. And, on the contrary, a “trial-centric” approach to find relevant patients for a given trial. In this study, we focus on the latter “trial-centric” task and adapt previous approaches for a better real-world applicability. Many previous studies have proven the theoretical feasibility and effectiveness of LLM-based eligibility assessments. To start with,³ introduced a framework leveraging LLMs to convert eligibility criteria into queryable expressions and match them against electronic health records (EHRs). Their approach demonstrated a patient-level recall varying between 67.3% and 76.8%. Though their focus was solely on the challenging area of oncology trials, it made it difficult to generalize to other therapeutic areas. Similarly,⁴ explored the use of LLMs, and particularly OpenAI’s InstructGPT (pre-ChatGPT chat assistant), to perform patient-trial matching on a synthetic patient dataset. They showed that while the AI standalone performance was sub-par (low criterion-level accuracy of 72% resulting in a patient-level recall of 50% only), it was still possible to use it as a copilot and thus achieve perfect patient-level recall and acceptable precision (71%) while reducing the number of criteria to be checked by experts by 90%. Then,⁵ introduced TrialGPT, a three-stage approach (retrieval, criterion-assessment, trial-level aggregation) focusing on the patient-centric task (i.e. not comparable to our trial-centric approach). The second stage, which it has in common with the trial-centric approach, leverages GPT-4 and achieves an 87% criterion-level accuracy on the SIGIR 2016 cohort.⁶

To evaluate the benefit of providing a rationale alongside the actual eligibility decision,⁷ assessed the interpretability of LLM-generated assessments. Clinicians reviewed the rationales generated by the LLM for each eligibility decision, demonstrating that the model can produce coherent explanations for 97% of its correct decisions and even 75% of its incorrect ones.

Despite promising feasibility, the above approaches fell short when applied in a real-world setting. This is mainly due to four challenges:

- *Lack of basic reasoning skills, limiting assessment quality:* Despite the improvements made to chat models (a.k.a. single-turn models, i.e. models optimized for fast conversational interactions with humans as opposed to reasoning models optimized for problem solving), even the most advanced ones, like GPT-4o, fail at some simple tasks. For example, date-related calculations or assessing some logical expressions remain a challenge for them.
- *Reliance on expert-refined criteria, limiting broad applicability:* Many of the aforementioned approaches^{3,7} rely on having experts ask the right questions or refine the eligibility criteria to make them less ambiguous. While it is interesting in the context of one particular trial, it is not compatible with a broadly applicable pre-screening tool. Having an LLM autonomously refine a criterion could also be an option, but as eligibility criteria typically undergo rigorous scrutiny to meet many constraints, modifying them could lead to unintended consequences, so we put aside this approach.
- *Lack of visual understanding, limiting medical record coverage:* Furthermore, despite the rise of EHRs, numerous institutions still rely heavily on handwritten notes⁸ showed that in a sample of 405 patient notes taken in 2020 and 2021 in the UK, 40% of notes were handwritten. Besides, a lot of data, even if digitized, remains better represented in a visual form, like tables, charts, and graphs. Passing all this data into a text-only LLM does not allow for fully leveraging the available information. Indeed, despite recent improvements, OCR and captioning pipelines are known to be expensive, time-consuming, and error-prone.⁹ showed that parsing + captioning of tables and graphs + embedding took more than 7 seconds per page while embedding the image of the page directly took less than half a second, while achieving better retrieval performance.
- *Lack of a generic integration with healthcare systems, preventing universal deployments:* There is no generic way to interface with research centers and hospitals’ information systems. Within the US alone, the EHR vendor market is highly fragmented¹⁰ showed that only 4.5% of Medicare beneficiaries had their meaningful-use (MU) medical records associated with a single vendor, and 19.8% of them had their MU

medical records stored in 8 or more vendors. And the problem is of course even more present at a global scale, e.g. in Europe where the top 15 EHRs accounted for only 58% of the market in 2022¹¹. The lack of such a generic integration is an obstacle to the extraction of structured data or even textual data in general. Finally, our experience shows that research centers—i.e., research-only sites—often lack EHR systems and instead rely on Clinical Trial Management Systems (CTMS) to store medical records and other patient data. This reliance further complicates efforts to develop a universal integration. Given that these freestanding research sites represent up to 25% of all U.S. research sites¹², it is crucial to design a generic tool capable of adapting to any system.

In this study, we present an integration-free, trial-centric pre-screening pipeline that directly processes unstructured medical records using reasoning-capable and vision-enabled LLMs. The approach combines multimodal retrieval with automated criterion assessment to handle complex eligibility rules, interpret visual and handwritten data without lossy text conversion, and operate across heterogeneous healthcare systems without custom integration. We validate the pipeline on both a public benchmark (n2c2 2018 cohort selection) and a multi-site real-world dataset, where it achieves state-of-the-art accuracy and substantially reduces the time required for eligibility review. These results demonstrate that the method is robust, scalable, and practical for real-world clinical trial patient matching.

Methods

Pipeline description

Our study builds on previous research by introducing key elements that make the process more applicable to the real world.

In particular, we leverage three recent innovations to address the aforementioned limitations of previous attempts.

- *Evolution of Vision-Language models (VLMs):* Recent LLMs have been adapted to accept image inputs. These models split each image into patches, embed them with a vision transformer, then route those embeddings through a cross-modal adapter, a module specifically trained to project vision tokens into the language model’s token-embedding space, e.g. CLIP¹³ or BLIP-2¹⁴. As a result, the model can then reason over layout, handwriting and figures without any OCR step, hence allowing our pipeline to fully leverage the information available in medical records and bypassing the need for heavy and error-prone OCR pipelines.
- *Visual retrieval:* Because of improvements made to the VLMs scene, extensive work has also been done to make retrieval tasks work in a fully visual manner. To address this need⁹ they introduced ColPali, a visual retriever, and while doing so they introduced ViDoRe, a benchmark to compare the performances of visual retrieval methods. Then,¹⁵ released a multimodal embedding model, relying on a single vector per image, making it much easier to scale, and outperforming all other models on the ViDoRe benchmark, making it a natural choice for us.
- *Reasoning model:* Finally, the rise of the new reasoning model paradigm, introduced by OpenAI’s o1, made it possible to evaluate more complex criteria and have the model succeed at assessing complex logical expressions or arithmetic tasks that chat models would fail at.

Figure 1 gives an overview of our whole pipeline, composed of three components: a trial preprocessing, a patient preprocessing, and a 2-step patient-trial matching. We describe below each component in detail.

Trial preprocessing. The goal of this step is to convert a block of free text eligibility criteria into a set of ready-to-assess and well-structured criteria. It is made of three independent LLM calls. All prompts can be found in Supplementary Method 3.

- *Split criteria:* When retrieving the eligibility criteria (i.e. from clinical-trials.gov), we get a free-text block. In order to evaluate these criteria one by one, we split this block into individual criteria.

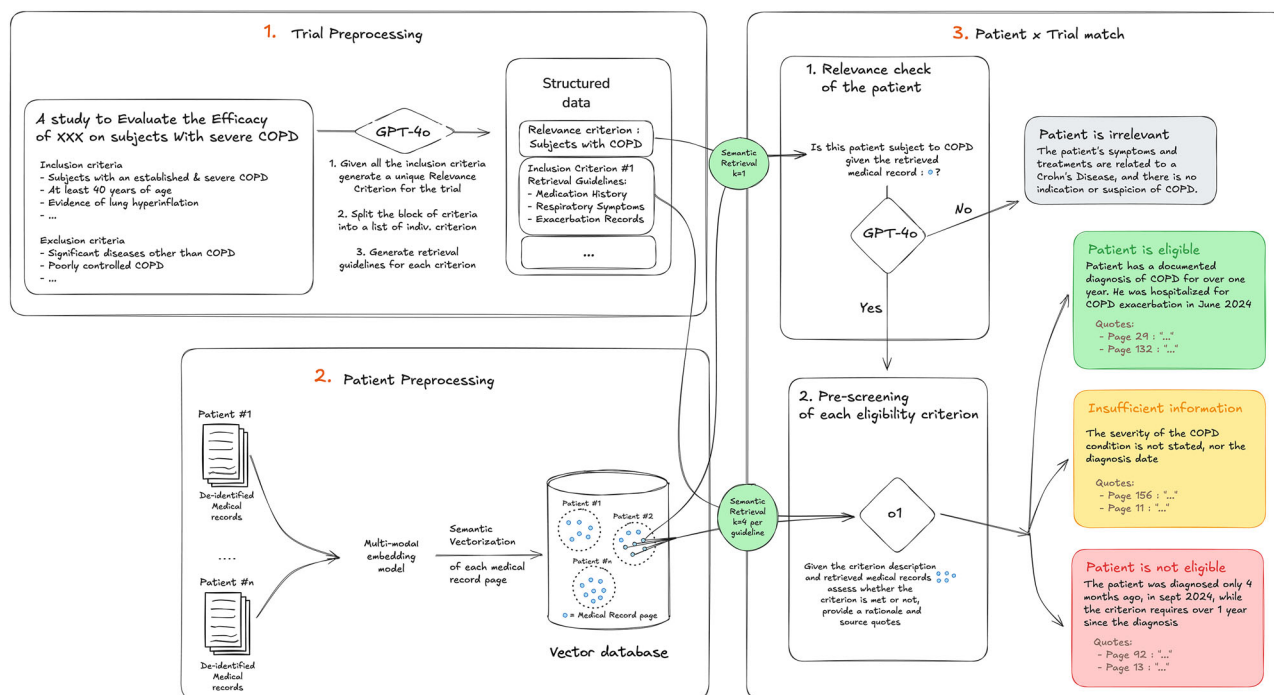


Fig. 1 | Our patient-trial matching pipeline

- **Generate relevance criterion:** Clinical sites databases usually contain many patients. For any given trial, most of them won't be relevant. E.g. a dermatology patient is irrelevant for a cardiovascular trial, so there is no need to pre-screen them for it, and we want to filter them upfront. For this purpose, we have an LLM generate a “relevance criterion”, typically containing the main condition targeted by the trial. It will be used later in the process to perform a low-cost relevance check for each trial x patient pair.
- **Generate retrieval guidelines:** We observed that, when confronted with large medical records, Clinical Researchers tend to make assumptions as to where they could find the relevant information. This is also recommended in ref. 16. In this fashion, we have an LLM generate such retrieval guidelines.

Patient preprocessing. We process uploaded files in the following way:

- **Split and de-identification:** We first split the PDF into individual images and anonymize them using Google Cloud DLP de-identification service to redact any patient-identifying information.
- **Embedding:** We send base64 encodings of these images to the multi-modal embedding model introduced by¹⁵.
- **Vector storing:** We store the resulting dense vectors into a vector database, for later semantic retrieval.

Patient x Trial matching. The patient-trial match is composed of two steps. Prompts can be found in Supplementary Method 4.

- **Relevance check:** In order not to waste resources on patients that are evidently not relevant to a given trial, we first make a low-cost relevance check. Using the relevance criterion generated in the section, we make a semantic retrieval of the most relevant page of the patient's medical record, i.e. we embed the relevance criterion with the same embedding model and execute a similarity search on our vector DB. We submit that page to GPT-4o to evaluate whether the patient meets the relevance criterion.
- **Assessment:** If the patient passed the relevance check, it is then assessed against each eligibility criterion of the trial. To do so, we use the guidelines generated in the section to semantically retrieve from the vector database the right parts of their medical records and submit

them to a multi-modal reasoning model (OpenAI's o1) to obtain an assessment (Eligible/Not Eligible/Insufficient information).

Throughout each phase of our pipeline, we opted for OpenAI's latest models due to their consistent top performance across multiple benchmarks. However, we did not conduct a comprehensive benchmarking evaluation of LLM providers.

User feedback

After the AI assessment is complete, users can confirm or rectify each criterion-level assessment and classify patients as “To screen” or “Not eligible”, hence giving us valuable feedback to measure the quality of our assessments.

Figure 2 shows what the assessments look like in our pre-screening tool, and how users can give feedback.

Datasets

We used two different datasets to evaluate our pipeline:

- **n2c2 public dataset:** We used the dataset from Track 1 of the 2018 n2c2 cohort selection benchmark, which is the largest publicly available clinical trial patient matching benchmark¹⁷. It was also used in ref. 7, so it makes it a natural choice to compare our performances. Though, as it's a text-only dataset, we converted the text to low-resolution images to test the visual aspect of our pipeline.
- **our own dataset based on in-app user feedback:** Following the release of our tool, we had it beta tested by 30 sites. We used their in-app feedback to evaluate our performance. The methodology to build this dataset is described below.

n2c2 public dataset. The 2018 n2c2 Clinical Trial Cohort Selection challenge dataset is composed of 288 unique diabetic patients, partitioned into a training set of 202 individuals and a test set of 86. Each patient has 2 to 5 clinical notes. These patients were originally drawn from a combined database at Massachusetts General Hospital and Brigham and Women's Hospital. Thirteen commonly employed inclusion criteria were used¹⁷. These criteria can be found in Supplementary Method 5. All patient data are provided in an unstructured format. To obtain the final labels, two

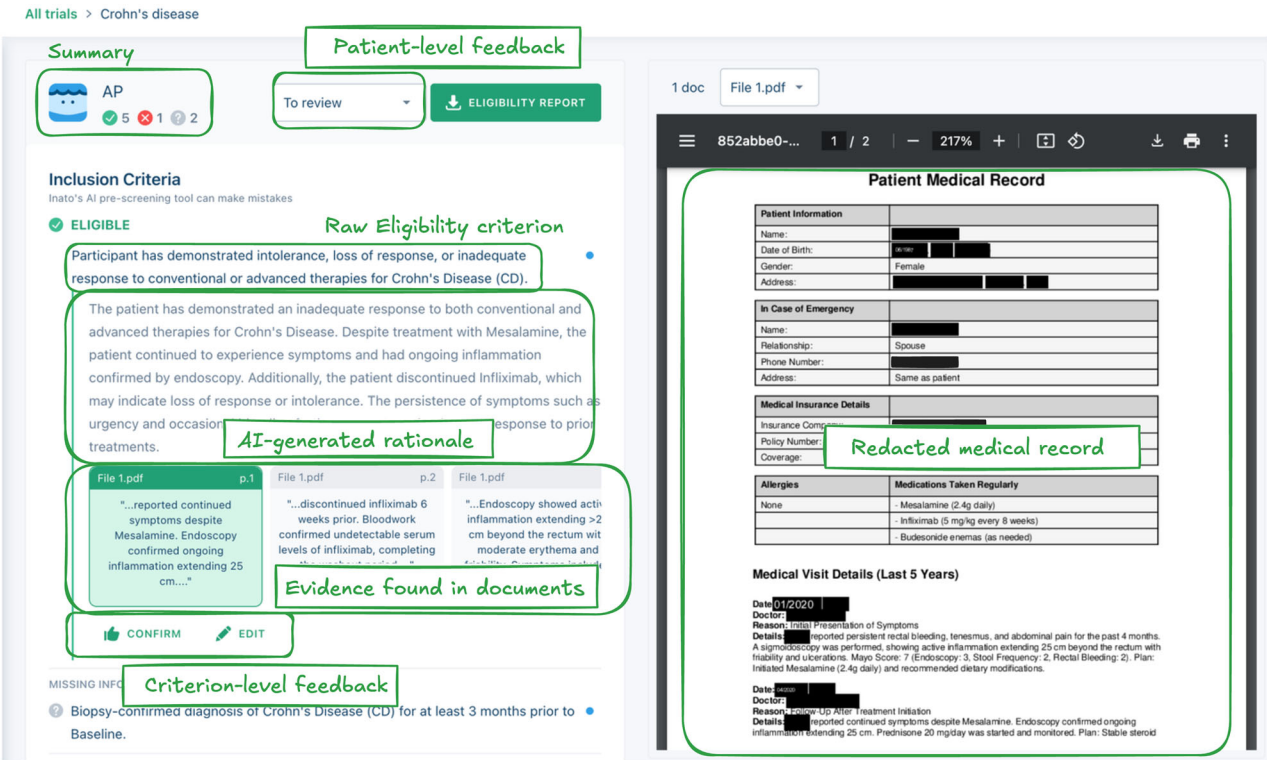


Fig. 2 | Screenshot of our prescreening tool - Patient pre-screened with rationale, source quotes and possibility to give feedback

medical experts independently annotated each patient’s status (whether the 13 criteria were met or not).

Due to the visual nature of our pipeline, we processed all clinical notes to convert them from text to image format. We chose a low resolution of 72 dots per inch to be conservative on the quality of images we would receive in real life.

Real-world dataset. As explained in the section, we gather feedback from our users in two different ways:

- via direct confirmation/rectification of each criterion-level assessment
- at the patient level as they classify them as “To be screened” or “Not eligible”

To build a gold-standard dataset for performance assessment, we used criterion-level reviews as ground truth whenever they were available. When only patient-level reviews were available, we relied on heuristics to deduce the true value of some criterion-level assessments. These heuristics are detailed in Supplementary Method 6.

Using this method, we obtained **7021** labeled patient-criterion pairs, distributed across 30 sites, 36 distinct trials, 485 patients and 650 eligibility criteria. We further describe them in the following subsections.

Sites. All 30 sites that beta tested our tool were located in the United States.

The single most active site represented 28% of the labeled criterion-level assessments, and the five most active represented 86% of them.

To qualify sites, we categorized them in the 6 categories reported in Table 1. Associated definitions can be found in Supplementary Method 11.

Evaluators. At each site, one person was in charge of reviewing the AI-generated assessments. We report in Table 2 their primary focus as they filled it in in their user profile.

Trials. The labels covered **36** different studies, evenly split between phase II and phase III trials. The focus of these trials was diverse, as shown in Table 3.

Table 1 | Summary of Research Site Types

Type	Count
Independent research site embedded in an outpatient clinic	11
Services company	6
Research center part of a site network	6
Independent research site	5
Outpatient clinic	1
Oncology care center	1

Table 2 | Summary of evaluators’s primary focus

Primary Focus	Count
Site Director	12
Study Coordinator	11
Other	3
Recruitment Specialist	3
Principal Investigator	1

The top-3 therapeutic areas (Gastroenterology, Pulmonology and Dermatology) accounted for 77% of all reviewed eligibility criteria.

Eligibility criteria. All eligibility criteria of these 36 trials were assessed by our tool, meaning that there was no hand-picking of the easy-to-assess criteria, resulting in a total of **650** criteria. To better qualify these criteria, we propose a 3-axis classification:

- **Domain:** Describes the general nature of the requirement (e.g., demographic detail, disease-specific factor, lab value, etc.).
- **Data Format:** Differentiates between criteria that can be captured as structured, numeric, or coded fields and those that rely on complex free-text.

- **Temporal Constraint:** Highlights whether the requirement applies within a particular time frame (e.g., “within 6 months” or “for at least 1 year”).

The classification of every criterion was done by GPT-4o. We present the results of this classification below.

We assign each criterion to its most relevant functional “domain”. These domains capture what the criterion is about. The prompt containing the definition of each domain and used to make this classification can be found in Supplementary Method 7. Table 4 gives the distribution of eligibility criteria per domain and an example criterion for each one of them.

We further classify criteria by the type of data they require to be assessed. The first group consists of criteria that can be evaluated using discrete or numeric data fields, codes, or straightforward yes/no inputs (we call these criteria *Structured*). And the other group contains all other criteria, i.e. those that require the interpretation of free-text medical notes or narratives (we call these criteria *Unstructured*). The prompt used to make this classification can be found in Supplementary Method 8. Table 5 gives the distribution of eligibility criteria per requested data format and an example criterion for each one of them.

Finally, we classify criteria depending on whether they mention a particular time frame (e.g., “within 6 months” or “for at least 1 year”). The prompt used to make this classification can be found in Supplementary Method 9. Table 6 gives the distribution of eligibility criteria per presence of temporal constraint and an example criterion for each one of them.

Medical records. In total, the medical records of **485** distinct patients were uploaded into our tool.

Table 3 | Therapeutic areas

Therapeutic Area	Trials	Labeled eligibility criteria
Pulmonology	11	1305
Gastroenterology	4	2950
Dermatology	3	1162
Cardiovascular	3	590
Endocrinology	3	306
Rheumatology	2	51
Psychiatry	2	80
Oncology	2	7
Hematology	2	7
Obesity	1	101
Pediatrics	1	51
Nephrology	1	420
Gynecology	1	9

Table 4 | Number of criteria assessments per domain (number of unique criteria)

Domain	Exclusion	Inclusion	Example
Comorbidity / Medical History	1224 (183)	248 (15)	History of dysplastic nevus syndrome
Demographic / Administrative	23 (2)	303 (32)	Participant must be 18 to 70 years of age inclusive
Disease or Condition Specific	126 (16)	587 (40)	Participant has an asthma diagnosis according to the GINA guidelines.
Lab or Biomarker	322 (35)	269 (25)	Absolute reticulocyte count above $100 \times 10^9/L$
Other Pragmatic	29 (5)	213 (15)	Willing and able to complete the scheduled study assessments
Functional Status	7 (4)	33 (10)	An ACQ-6 score of 1.5 or above
Prior / Concomitant Treatments	1326 (142)	277 (24)	Immunoglobulin or blood products within 4 weeks of Visit 1
Safety or Risk	1086 (96)	6 (5)	Primary or secondary immunodeficiency

Each patient was on average associated with 1.6 medical records (i.e. PDF files): most patients (69% of them) had only one medical record, but some of them had up to 9 medical records associated.

On average, medical records were **28** pages long. 50% of them were 12 pages or more and 10% of them were 73 pages or more.

To better qualify the content of medical records, we used the following categorization:

- **Administrative Documents:** These include forms and records capturing patient identification, insurance details, and legal consent (e.g., registration forms and HIPAA notices).
- **Clinical Notes:** This category encompasses encounter documentation such as History and Physical reports, progress notes, consultation notes, and discharge summaries, along with transition of care documents.
- **Diagnostic Reports:** These consist of laboratory, imaging, and pathology reports that present clinical test results and diagnostic findings (e.g., CBCs, CT scans, biopsy reports).
- **Procedural & Treatment Documents:** This group includes procedure reports (e.g., operative, endoscopy, and interventional reports) as well as medication orders and treatment plans.

To understand the distribution of our dataset, we randomly sampled 100 medical records, broke them down into 2947 individual images and used an LLM to classify each page into one of these 4 categories. The result of this categorization is shown in Table 7.

Statistics and reproducibility

Analyses were performed at the criterion level, where each patient-criterion pair was treated as an independent sample. For the n2c2 benchmark dataset, this resulted in 2366 assessments across 288 patients and 13 criteria. For the real-world dataset, 7021 assessments were obtained from 485 patients across 36 trials and 650 eligibility criteria, contributed by 30 research sites. Replicates are therefore defined as independent patient-criterion assessments, including multiple patients per site and multiple criteria per patient. Performance metrics (accuracy, precision, recall, F1-score) were computed using standard definitions. No additional statistical resampling was applied, and results are reported as observed on the full datasets, which provide a large and diverse sample supporting the robustness and reproducibility of findings.

Table 5 | Number of criteria assessments per requested data format (number of unique criteria)

Data required	Exclusion	Inclusion	Example
Structured	967 (105)	981 (89)	Have a BMI 23.0 kg/m ² at Visit 1
Unstructured	3185 (379)	955 (77)	Are currently receiving treatment for diabetic retinopathy

Table 6 | Number of criteria assessments per temporal constraint (number of unique criteria)

Temporal constraint	Exclusion	Inclusion	Example
Yes	1905 (211)	660 (57)	Documented physician diagnosis of asthma for 12 months or more prior to Screening
No	2247 (273)	1276 (109)	Extensive-stage SCLC

Table 7 | Distribution of Medical Records types

Type	Pages	Medical records were present
Clinical Notes	1519	85
Diagnostic Reports	785	69
Procedural and Treatment Documents	467	69
Administrative Documents	176	51
Total	2947	

Example: Out of 2947 pages analyzed, 785 of them were diagnostic reports. This type of content was present in 69 of the 100 medical records analyzed.

Table 8 | Criterion-level classification report of our visual o1-based pipeline on the n2c2 dataset converted in low-resolution images, leveraging all available clinical notes

Class	Precision	Recall	F1-score	Sample size
met	0.92	0.93	0.93	1041
unmet	0.94	0.93	0.93	1325
Accuracy			0.93	2366
Macro avg	0.93	0.93	0.93	2366
Weighted avg	0.93	0.93	0.93	2366

Table 9 | Criterion-level classification report of our visual o1-based pipeline on our real-world dataset, leveraging the top-3 clinical notes, with guidelines activated

Class	Precision	Recall	F1-score	Sample size
met	0.72	0.76	0.74	1683
unknown	0.97	0.99	0.98	1999
unmet	0.88	0.85	0.86	3339
Accuracy			0.87	7021
Macro avg	0.86	0.86	0.86	7021
Weighted avg	0.86	0.87	0.87	7021

Table 10 | Subset classification report on inclusion criteria only - Real-World dataset

Class	Precision	Recall	F1-score	Sample size
met	1.00	0.88	0.93	798
unknown	0.94	0.94	0.96	518
unmet	0.86	0.97	0.91	562
Accuracy			0.93	1878
Macro avg	0.93	0.94	0.93	1878
Weighted avg	0.94	0.93	0.93	1878

Ethics

This study used only anonymized patient data that had been de-identified under the HIPAA Safe Harbor method, and therefore did not require Institutional Review Board approval.

Table 11 | Subset classification report on exclusion criteria only - Real World dataset

Class	Precision	Recall	F1-score	Sample size
met	0.59	0.99	0.74	411
unknown	0.63	0.99	0.77	1327
unmet	1.00	0.88	0.93	2360
Accuracy			0.86	4098
Macro avg	0.86	0.95	0.81	4098
Weighted avg	0.95	0.86	0.90	4098

Results

Results on the n2c2 dataset

Table 8 showcases the results of our pipeline on the n2c2 dataset.

Results on our real-world dataset

Table 9 showcases the results of our pipeline on real-world data, as detailed in the above section. Tables 10 and 11, respectively, give results for inclusion and exclusion criteria.

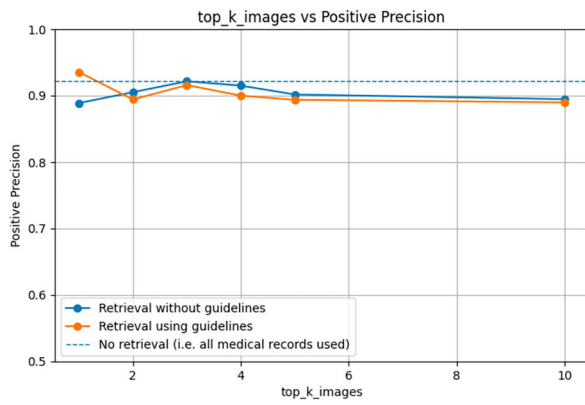
Effect of retrieval and guidelines. To measure the effect of retrieval and guidelines, we first measure the effect of having a semantic retrieval steps rather than using all clinical notes, on the n2c2 dataset. Additionally, we measure the performance of guidelines vs no guidelines.

Figure 3 shows that recall improves as we use more images. But interestingly, the precision is more or less always the same, even for a low number of images retrieved. This is likely explained by the fact that the n2c2 dataset is composed exclusively of inclusion criteria, and the LLM never considers that criteria are met if there is no tangible proof, resulting in a low number of false positives. Besides, we notice that none of the retrieval strategies are good enough to match the performance of the all-notes run. This was already observed by Wornow et al.⁷

The use of guidelines seems beneficial for the recall, but the comparison is unfair because when using guidelines, K images are retrieved for **each** guideline, and there are several guidelines per criterion, so on average, more images are used for each criterion assessment. To visualize the real effect of guidelines, we plot the recall according to the average number of images used by each retrieval strategy. Figure 4 shows that the benefit is actually almost inexistent or even detrimental. We were surprised by this result as having a research guideline is a clearly beneficial technique for manual patient review. Because we did not have a labeled retrieval dataset, we could not investigate further, but we believe there is still significant room for improvement in this area—particularly in the use of dynamic retrieval, where the model could successively retrieve pages as it reasons, rather than performing a single upfront retrieval step.

Accuracy by criterion domain. In Table 12 we share the accuracy for each domain, as described in Supplementary Method 7. There is no large variability between domains.

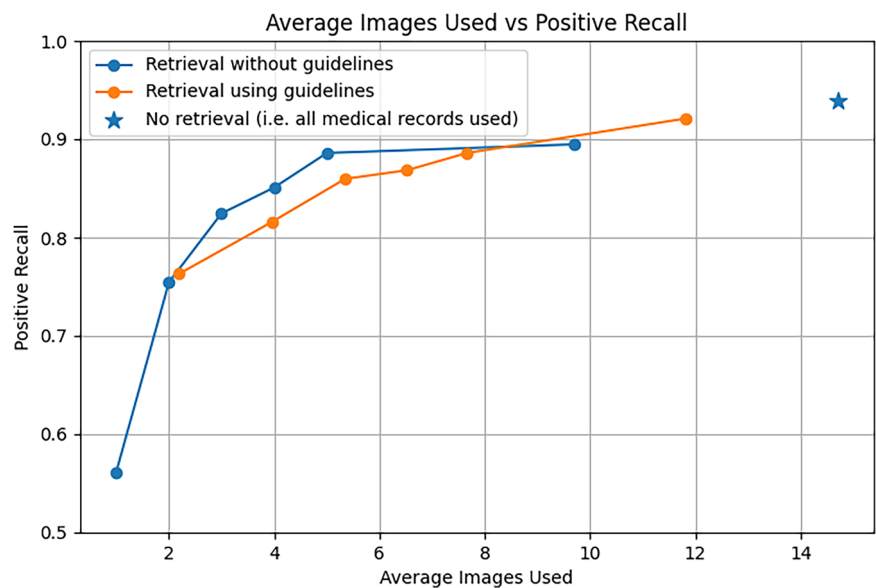
Inference efficiency metrics. When assessing criteria using all clinical notes, our pipeline took on average 24.8s per criterion, with an interquartile range of 18.5s–28.2s. This average fell to 19.0s per criterion when using only the top-3 clinical notes for each retrieval guideline.



(a) Precision



(b) Recall

Fig. 3 | Precision and Recall per average images used for each retrieval strategy**Fig. 4** | Recall per average images used for each retrieval strategy**Table 12** | Accuracy across domains

Domain	Accuracy	Samples
Comorbidity or Medical History	0.86	1792
Demographic / Administrative	0.89	352
Disease or Condition Specific	0.88	886
Lab or Biomarker	0.87	747
Other Pragmatic	0.97	229
Performance or Functional Status	0.84	32
Prior or Concomitant Treatments	0.86	1933
Safety or Risk	0.87	1112

As for the inference cost, our pipeline yielded a criterion assessment for an average cost of \$0.15. That cost decreased to \$0.09 when using only the top-3 clinical notes for each retrieval guideline.

User review efficiency metric. As our paper focuses on enabling real-world productivity gains, we measured the time users took to review a complete patient-trial pair.

The median time observed to review a patient was **5.5 min**, the mean was around **9 min**, and the interquartile range spanned from 3 to 11 minutes. Heuristics used to obtain these metrics are described in Supplementary Method 10.

Discussion

By comparing our results shown in Table 8 with previous attempts made with GPT-4 (Precision **0.91**, Recall **0.92** on the positive class⁷), we demonstrate that our o1 visual pipeline outperforms the previous SOTA despite relying on a degraded version of the dataset (resulting from the conversion of text clinical notes into low-resolution images). This advantage becomes especially clear when excluding previous approaches that relied on refined criteria (in which case GPT-4 achieved a precision of 0.89 and a recall of 0.74 on the positive class, as shown in ref. 7), an aspect we consider non-scalable, as explained in the introduction. Table 9 presents the overall performance achieved by our pipeline on our real-world dataset, showcasing an overall accuracy of 87%, slightly below what we witnessed on the n2c2 public dataset.

To shed some more light and understand why the accuracy observed in our real-world experiment is lower than the one obtained on the public dataset, we present separately the classification reports of inclusion

(Table 10) and exclusion criteria (Table 11). This way, we observe that the majority of mislabeled samples (>80%) are coming from patients that we classified as Ineligible either due to an unmet inclusion criterion or a met exclusion criterion. This, along with user feedback analysis, possibly shows that CRCs using the tool are reluctant to exclude a patient without doing further investigations. Both the met inclusion criteria and unmet exclusion criteria categories achieved a very high precision of 1.00.

Finally, in the section, we show that the time required for prescreening using our tool is over 80% more efficient than traditional manual reviews, as noted in ref. 2. However, this comparison is limited to the prescreening phase only, and does not reflect the complete screening process, which also includes patient interactions.

Conclusions

This study presents a multimodal AI pipeline that automates the matching of patients to clinical trials directly from unprocessed medical records. By combining reasoning models with visual understanding and efficient document retrieval, our approach achieved state-of-the-art performance on a public benchmark and demonstrated robust accuracy in a real-world setting across diverse sites and trials. Importantly, the tool reduced prescreening time by more than 80% compared with traditional manual review, allowing coordinators to confirm eligibility in minutes rather than hours.

Despite these gains, two challenges remain. First, user feedback shows that research staff are often reluctant to exclude patients without further checks, highlighting the difficulty of calibrating between firm and uncertain assessments when records are incomplete. Second, while criterion-level assessments are reliable, aggregating them into a single patient-level recommendation is more complex. We explored several approaches around absolute count and percentage of failing/succeeding criteria, but none of them offered an acceptable patient-level precision/recall trade-off.

Overall, this work demonstrates that an integration-free, multimodal AI system can meaningfully accelerate trial recruitment while maintaining high-quality eligibility assessments. Addressing the remaining challenges around uncertainty calibration and patient-level aggregation will further enhance its clinical utility.

Data availability

The 2018 n2c2 cohort-selection dataset used in this study was obtained from the Harvard medical school portal (<https://n2c2.dbmi.hms.harvard.edu/data-sets>) prior to its suspension. As of 8 October 2025, the portal is not accepting new access requests; consequently, these source data are not currently available to new applicants. As for the real-world dataset, we are unable to share the criteria due to Non-Disclosure Agreements covering the trial protocols. Enquiries can be sent to anatole.callies@inato.com.

Code availability

All the required code to replicate our results on the public n2c2 dataset is publicly available at https://github.com/inato/ai_trial_eligibility_assessment.

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Author contributions

Anatole Callies and Quentin Bodinier contributed equally.

Competing interests

The authors are employees or administrators of Inato, a company that develops products evaluated in this study.

Additional information

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Correspondence and requests for materials should be addressed to Anatole Callies.

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